Deriving the equations of motion

AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.9 | Sections 2.9

Goal: Derive the four suvat equations from a velocity–time graph.

01 The big picture

For uniform acceleration the v–t graph is a straight line from u to v: its gradient is a and the area under it is s.

Those give the first two equations; algebra gives the other two.

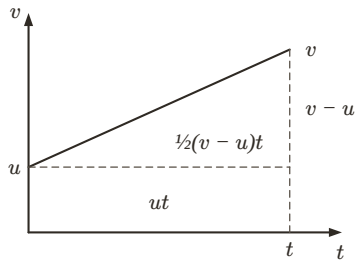
Uniformly accelerated motion [paraphrase]
Constant acceleration in a straight line.

02 Deriving the equations

D1 From the v-t graph

Velocity rises uniformly from u to v in time t.

Fig. 1 · The area under a uniform v-t line



Rectangle ut plus triangle $\frac{1}{2}(v-u)t$; the gradient is a.

1. Gradient = acceleration: $a = \dfrac{v-u}{t} \Rightarrow v = u + at$
2. Area = rectangle + triangle:
$$s = ut + \tfrac{1}{2}(v-u)t = \tfrac{1}{2}(u+v)\,t$$
3. Put $v - u = at$ into the area: $s = ut + \frac{1}{2}at^2$
4. Put $t = (v-u)/a$ into the area:
$$s = \frac{(v+u)(v-u)}{2a} \Rightarrow v^2 = u^2 + 2as$$

RESULT

$$v = u + at, \qquad s = \tfrac{1}{2}(u+v)\,t,$$
$$s = ut + \tfrac{1}{2}at^2, \quad v^2 = u^2 + 2as$$

03 Sample problems with solutions

Sample 1 · Area check

An object accelerates uniformly from 4.0 m s^{-1} to 10 m s^{-1} in 3.0 s. Find s from the area, then check it with $s = ut + \frac{1}{2}at^2$.

Step 1. Area (trapezium):
$$s = \tfrac{1}{2}(4.0 + 10)(3.0) = 21 \text{ m}$$

Step 2. Check: $a = (10 - 4.0)/3.0 = 2.0$ m s^{-2}, so
$$s = 4.0(3.0) + \tfrac{1}{2}(2.0)(3.0)^2 = 12 + 9.0 = 21 \text{ m} \checkmark$$

Answer: $s = 21$ m both ways

04 Practice problems

P1 Which feature of a v–t graph gives $v = u + at$?
Answer: The gradient

P2 Which suvat equation has no t, and how is it derived?
Answer: $v^2 = u^2 + 2as$ (eliminate t)

P3 In $s = ut + \frac{1}{2}at^2$, which part of the graph area is ut?
Answer: The rectangle under u

P4 Find s from the area: $u = 0$, $v = 12$ m s^{-1}, $t = 5.0$ s.
Answer: 30 m

05 Other useful information

Common mistakes

✗ **Wrong:** Use suvat for any motion.
✓ **Right:** Only for constant acceleration.

Exam tips

- Forgot an equation? Sketch the v–t trapezium.

What the syllabus asks

- 9702 topic 2: derive the suvat equations. Coursebook 2.9.

Explore: Practice questions for this lesson